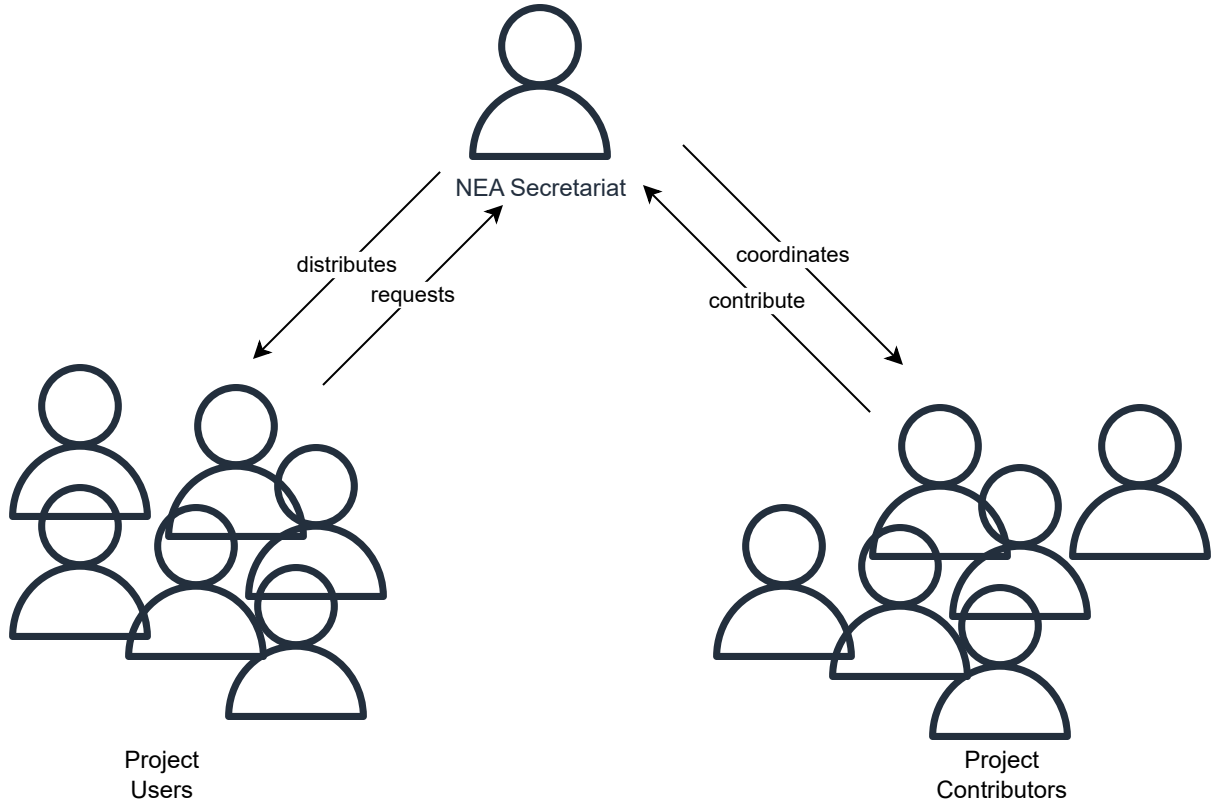
NEA
NUCLEAR ENERGY AGENCY

Strengthening Collaboration and Reproducibility in the Nuclear Data Community

Challenges and NEA Initiatives

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CM on Follow-up Developments in Nuclear Data Retrieval, Dissemination, and Data Portal
https://doi.org/10.5281/zenodo.23667633

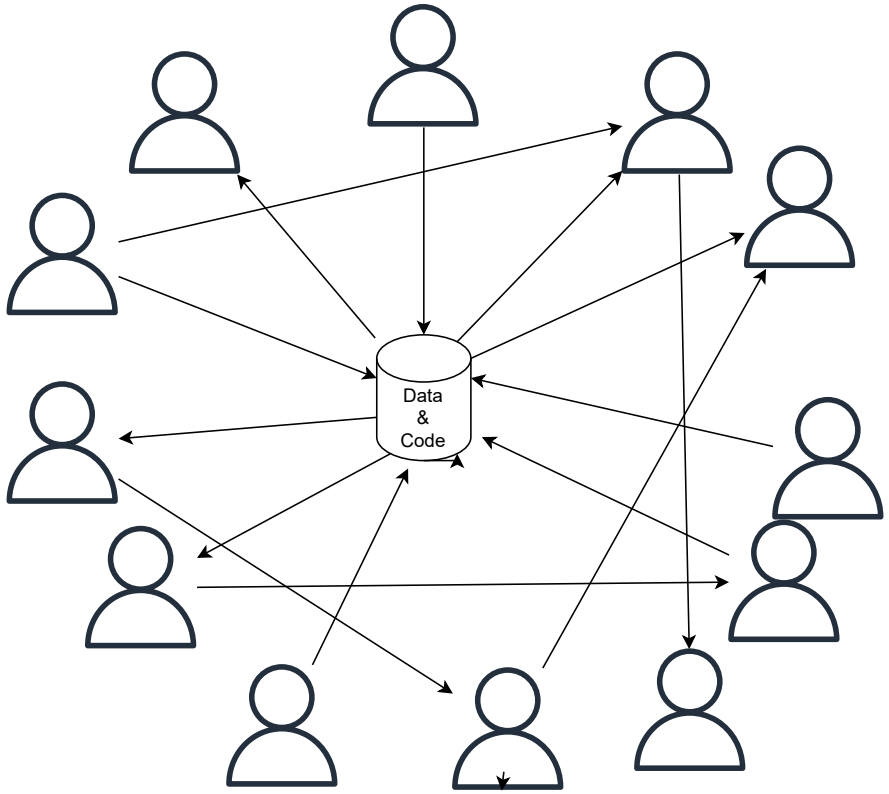
NEA Coordinated Groups - Collaboration Mode



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Participating in international collaborations



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NEA Requirements for Collaborations

- Transparent contribution data flow & contribution attribution -> **Version Control**
- Project community takes ownership of products -> **Full community control**
- Persistence of release versions -> **PIDs, Datacite integration**

Good to have

- Common standards between projects
  - Project management -> **NEA GitLab common practices**
  - Data structures -> **Shared modules across projects**
- Integration of ontology data

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Interoperability & Modularization

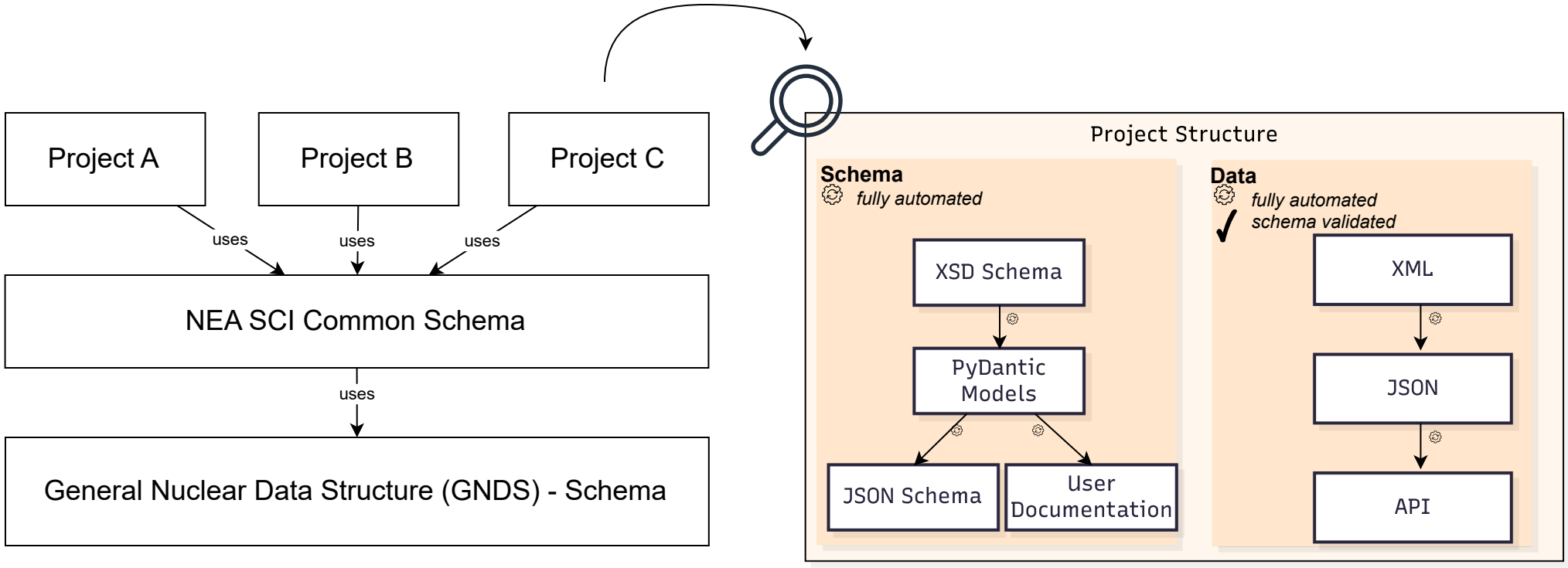
Modernisation of NEA Nuclear Science (SCI) projects

- Migration to structured data formats
- Format validation through schemas
- Modularisation of schemas
- Enables ontology integration

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NEA Nuclear Science Projects Vision



Example: Spend Fuel Composition Database SFCOMPO

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NEA Nuclear Science Projects

Unlocked project features

- Git-based version control
- GitLab-based community collaboration
- Transparent & automated
  - conversion
  - validation
  - deployment
- Direct user feedback & integration of user community

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Opening of data standards and facilitated data contribution

Exfor Submission Templates

Opening pathway for data contribution while following best-practices for data management

- Graphical User Submisison Template based on JSON Forms
  - Graphical submission form
  - Immediate JSON schema validation
  - Hosted as GitLab Page

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Exfor Submission Templates

Challenge: No direct submission pathway yet (authentication, backend required)

Need to find optimum between user friendlyness and systematic version control and system security

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Persistend Identifiers, Rich Metadata and References

NEA Data Platform / InvenioRDM

- Persistent identifiers for long-term referencing of research resources
- Integration of (data and code) publications into the DataCite universe
- Accessible via standard interfaces -> indexing
- Findable via DataCite metadata and relation to other resources

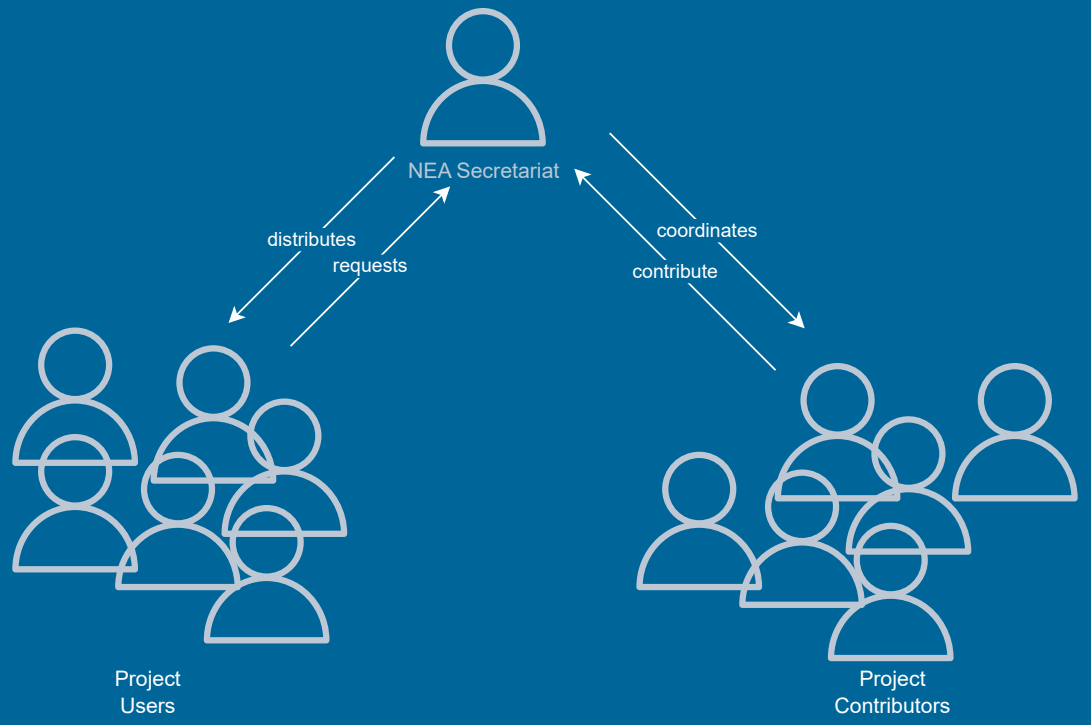
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Discussion

Other approaches to foster community involvement?

- How to lower adoption barrier Git / GitLab for non technical audiences?
- How to foster sustainable community engagement?
- How to facilitate project ownership?

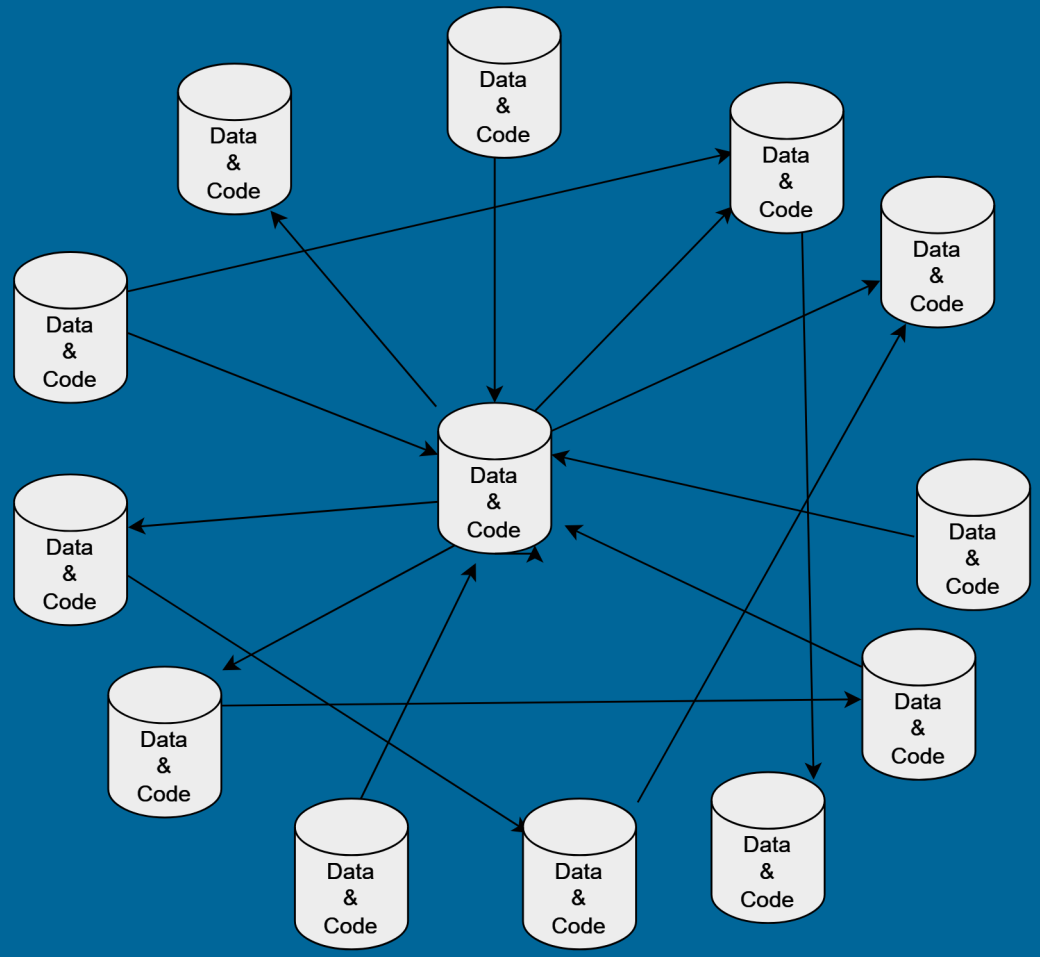


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How to improve …

- Interoperability of databases
  - Dataexplorer, Medusal, next generation JANIS, Data Portal
- Dataset relations & findability
  - PIDs, Ontologies
- Dataset access
  - Open-sourcing legacy datasets



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